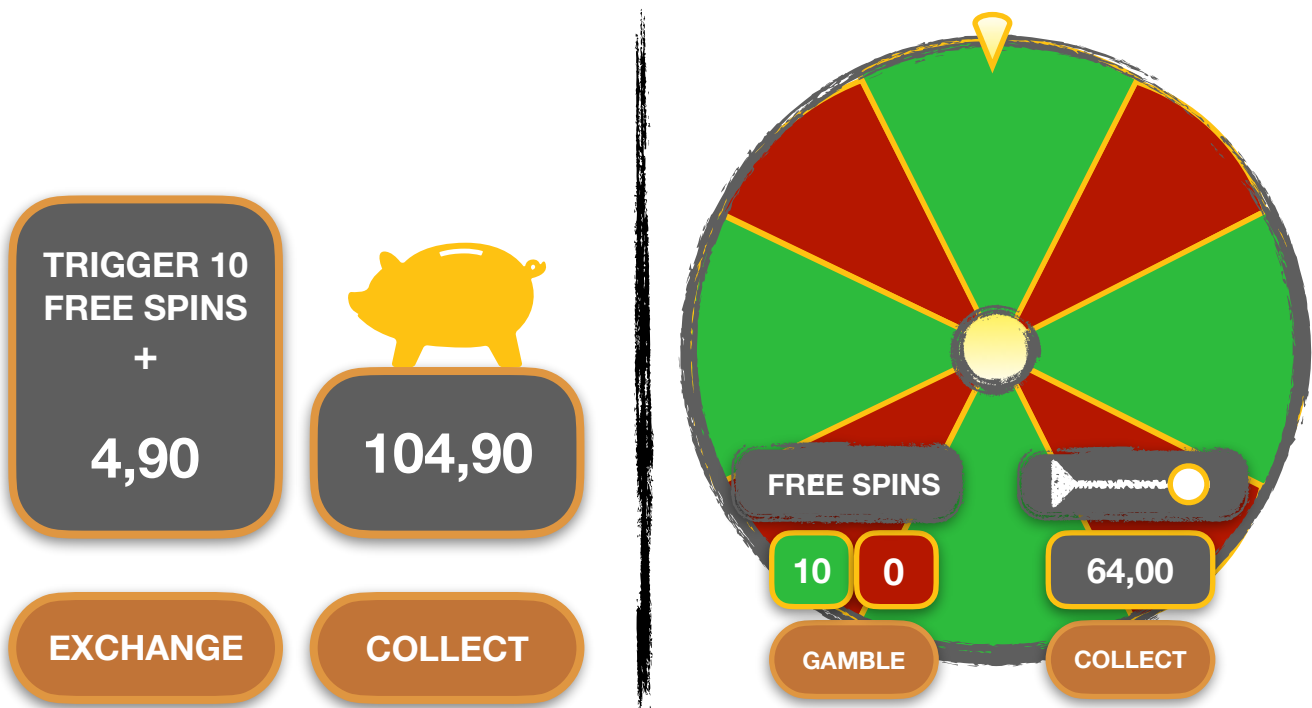


# WIN EXCHANGE Feature

## Overview

The **WIN EXCHANGE** feature introduces a controlled and player-driven mechanism that bridges the gap between **base game wins** and **access to the main bonus game**. It is designed to improve player engagement, preserve the mathematical integrity of rare high-value bonuses, and offer configurable operator flexibility without relying on aggressive mechanics such as Bonus Buy or Ante Bet.

The core idea is simple: **players can exchange certain base-game wins for a chance to trigger the main bonus feature**, under carefully balanced probability conditions.



When you win  $100 \times$  your stake or more, you have the option to change  $100 \times$  your stake for triggering Free Spins bonus while collecting any remaining winnings.

When you win between  $30x$  and  $100x$  your stake, you have the option to exchange the entire win amount (or part of it, not lower than  $20 \times$  for example) for the chance to be awarded bonus B. The chance of winning the gamble is shown in green on a wheel.

## Problem Statement

In current slot designs, there is a structural tension between:

- **Bonus frequency** (often constrained to  $\sim 1/150$  to  $1/300$  for operational reasons)
- **Bonus quality** (higher volatility and richer payout structures require lower frequency)
- **Player engagement** (players often prefer more frequent access to bonus gameplay)

When bonus frequency is forced higher, the bonus must be diluted in value. When it is too rare, engagement may drop.

WIN EXCHANGE aims to resolve this contradiction in a mathematically controlled and optional way.

## Core Concept

The WIN EXCHANGE feature activates when a player achieves a qualifying win in the base game.

Two thresholds are defined:

### 1. High-tier wins (e.g. $\geq 100x$ stake)

- Player is offered the option to exchange part or all of the win for a **chance to enter the main bonus (Bonus B)**.
- Any non-exchanged portion is retained as normal cash win.

### 2. Mid-tier wins (e.g. $30x - 100x$ stake)

- Player may exchange the entire win for a **probabilistic entry into the bonus feature**.
- No partial cash-out is retained in this case.

The exchange is always optional, preserving player agency.

## Probabilistic Balancing Model

The probability of entering the bonus is dynamically adjusted based on:

- The size of the exchanged win ( $W$ )
- The theoretical mean value of the bonus ( $M$ ), derived from PAR simulation data
- Predefined RTP and volatility constraints

A simplified conceptual model:

- Higher exchanged wins → higher probability of bonus entry
- Lower exchanged wins → lower probability of bonus entry

The system ensures:

- No exploitable deterministic strategy
- No RTP inflation beyond design limits
- Preservation of bonus rarity and value perception

Random outcome determination is handled via standard RNG (e.g. 1–1000 scale), ensuring precision up to two decimal probability points.

## Game Design Intent

The WIN EXCHANGE system allows:

- **Rare but high-quality bonus design (e.g. 1/400–1/600)** to remain intact
- **Flexible player-driven access** to bonus gameplay
- **Retention of volatility structure** in the base game
- **Configurable engagement tuning** without changing core math models

This creates a dual-layer system:

- Base game remains statistically stable and high quality
- Bonus entry becomes partially player-influenced without being fully deterministic

## Implementation Concept

To ensure simplicity of integration, the feature is exposed via a deterministic configuration model.

The game defines:

- Fixed number of wheel segments (N)
- Angular segmentation of a full circle ( $2\pi$  radians or  $360^\circ$ )
- Dynamic assignment of:
  - **Winning probability arc**
  - **Losing probability arc**

The backend provides a DTO containing:

[github.com/andy489](https://github.com/andy489)

- Segment count
- Segment sizes (degrees or radians)
- Probability distribution parameters

The frontend renders a wheel where:

- Green sectors represent successful bonus entry outcomes
- Red sectors represent unsuccessful outcomes
- Sector sizes are proportional to calculated probabilities

This ensures a fully transparent and deterministic rendering layer while maintaining RNG fairness on the backend.

```
{
  "wheel": {
    "segmentCount": 12,
    "segments": [
      {
        "type": "WIN",
        "probability": 0.730,
        "angleDegrees": 262.8
      },
      {
        "type": "LOSE",
        "probability": 0.270,
        "angleDegrees": 97.2
      }
    ]
  }
}
```

## Mathematical Model (How probabilities are calculated)

**Step 1** – define your system, compute the win multiplier and the bonus mean (by Par Sheet, i.e. empirically with simulation).

We define RNG range: 1–1000

$$\begin{cases} \omega = \text{stake multiplier from base game} \\ \mu_X = \text{mean stake multiplier for bonus X} \end{cases}$$

## Step 2 – Compute probability (simplest model)

$$P_{ex}(\omega) = \frac{\omega}{\mu_X}$$

For example:

$$\mu_X = 100 \times \text{stake}$$

$$\omega = 73 \times \text{stake}$$

$$P_{ex}(\omega) = \frac{\omega}{\mu_X} = \frac{73}{100} = 0.730$$

So probability of entering a bonus with mean stake multiplier of  $\mu_X = 100 \times \text{stake}$  if the player exchanges a win amount of  $\omega = 73 \times \text{stake}$  is **73%**.

## Step 3 – Construct DTO.

We can actually take this further and evolve the feature and **instead of fixed exchange probability, you now introduce a player-controlled cost-to-enter-bonus**.

### Request DTO (what frontend sends)

```
{
  "gameId": "WIN_EXCHANGE_X",
  "sessionId": "abc123",
  "spinResult": {
    "stake": 1.0,
    "winMultiplier": 84.3
  },
  "exchangeRequest": {
    "selectedCostMultiplier": 22.0
  }
}
```

### Backend Response DTO

```
{
  "exchangeDecision": {
    "approved": true,
    "appliedCostMultiplier": 22.0,
    "remainingWinMultiplier": 62.3
  },
  "bonusEntryModel": {
    "type": "PROBABILISTIC_BONUS",
    "entryProbability": 0.220,
  },
  "constraints": {
    "minCost": 20.0,
    "maxAllowedByWin": 84.3
  },
  "rng": {
    "range": [1, 1000],
    "roll": 114,
    "result": "SUCCESS"
  }
}
```

# RISK YOUR WIN SPIN FOR BONUS X



## 1. YOU WIN!

You've won with a stake multiplier.



## 2. EXCHANGE

Exchange part or all of your win for a chance to enter the main bonus.



## 3. SPIN

Spin the wheel. Will fortune be on your side?



### HOW MUCH WILL YOU EXCHANGE?

KEEP MORE  
Lower chance



EXCHANGE: 0%



100%



RISK MORE  
Higher chance

RISK IT. SPIN IT. WIN BIG.

BONUS X

## Design Advantages

### 1. Preserves bonus quality

Bonuses remain rare and high-value, maintaining strong player anticipation.

### 2. Adds controlled interactivity

Players feel agency over bonus entry without breaking mathematical balance and without injecting best game strategy.

### 3. Avoids aggressive monetization mechanics

Unlike Bonus Buy, no direct purchase is involved.

#### 4. Maintains operator flexibility

Frequency and thresholds can be tuned without altering core RTP logic.

#### 5. Scalable framework

Can be extended to multiple bonus types or layered feature systems.

### Compliance & Regulatory Considerations

Before deployment, the following must be evaluated:

- Whether probabilistic win-to-feature exchange is classified as gambling augmentation in target jurisdictions
- Whether dynamic conversion of winnings into feature chances affects RTP disclosure requirements
- Whether similar mechanics are already patented or restricted by existing providers

Legal review is recommended prior to implementation in regulated markets.

### Conclusion

WIN EXCHANGE introduces a **hybrid mechanic between base-game wins and bonus entry probability**, allowing operators to maintain rare, high-quality bonus systems while giving players meaningful control over when they access them.

It is a mathematically controlled, configurable, and scalable framework designed to enhance engagement without compromising game integrity or volatility structure.